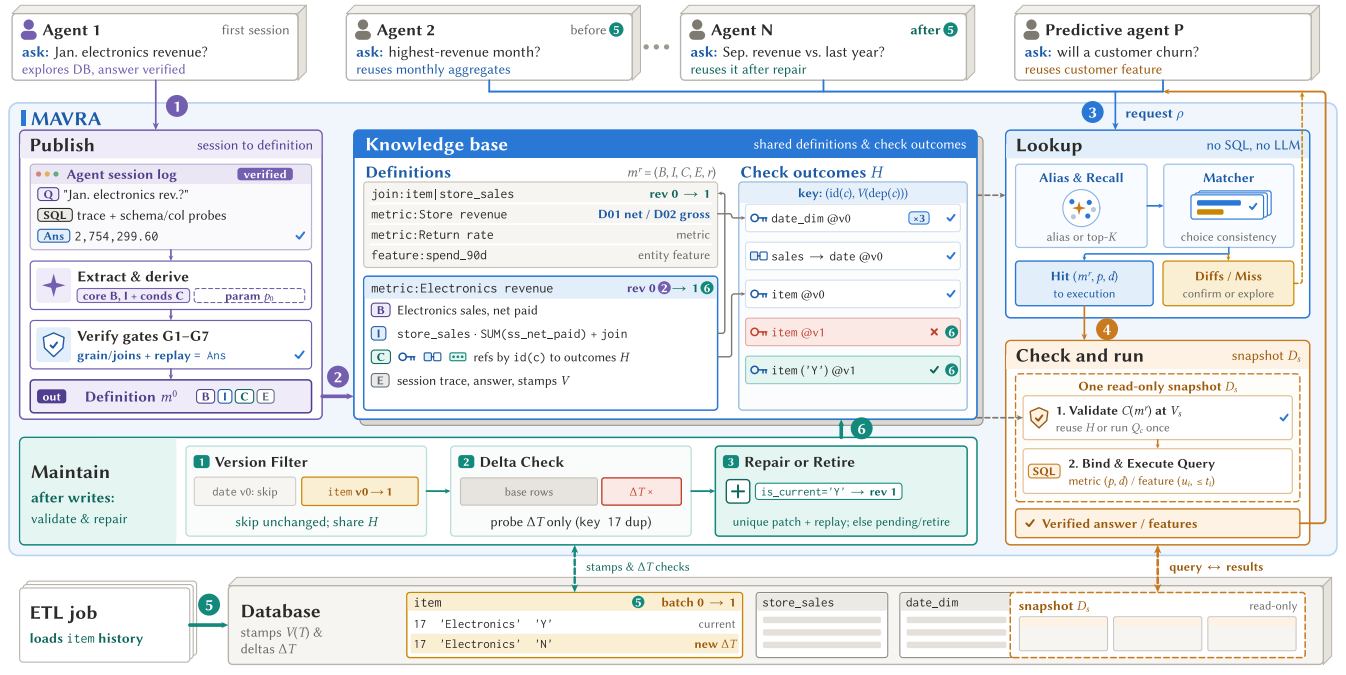

Agent 1
first session
ask: Jan. electronics revenue?
explores DB, answer verified
Agent 2
before 5
ask: highest-revenue month?
reuses monthly aggregates
Agent N
after 5
ask: Sep. revenue vs. last year?
reuses it after repair
Predictive agent P
ask: will a customer churn?
reuses customer feature
MAVRA
1
request ρ
3
Publish
session to definition
Agent session log
verified
Q "Jan. electronics rev.?"
SQL trace + schema/col probes
Ans 2,754,299.60
Extract & derive
core B, I + conds C
param p0
Verify gates G1–G7
grain/joins + replay = Ans
out Definition m0
B I C E
2
Knowledge base
shared definitions & check outcomes
Definitions
m' = (B, I, C, E, r)
join:item|store_sales
rev 0 → 1
metric:Store revenue
D01 net / D02 gross
metric:Return rate
metric
feature:spend_90d
entity feature
metric:Electronics revenue
rev 0 2 → 1 6
B Electronics sales, net paid
I store_sales · SUM(ss_net_paid) + join
C refs by id(c) to outcomes H
E session trace, answer, stamps V
Check outcomes H
key: (id(c), V(dep(c)))
date_dim @v0
×3
sales → date @v0
item @v0
item @v1
6
item ('Y') @v1
6
Lookup
no SQL, no LLM
Alias & Recall
alias or top-K
Matcher
choice consistency
Hit (m', p, d)
to execution
Diffs / Miss
confirm or explore
4
Check and run
snapshot Ds
One read-only snapshot Ds
1. Validate C(m') at Vs
reuse H or run Qc once
SQL 2. Bind & Execute Query
metric (p, d) / feature (u0, ≤ t1)
Verified answer / features
6
Maintain
after writes:
validate & repair
1 Version Filter
date v0: skip
item v0 → 1
skip unchanged; share H
2 Delta Check
base rows
ΔT ×
probe ΔT only (key 17 dup)
3 Repair or Retire
is_current='Y' → rev 1
unique patch + replay; else pending/retire
stamps & ΔT checks
query ↔ results
ETL job
loads item history
5
Database
stamps V(T) &
deltas ΔT
item
5 batch 0 → 1
17 'Electronics' 'Y'
current
17 'Electronics' 'N'
new ΔT
store_sales
date_dim
snapshot Ds
read-only